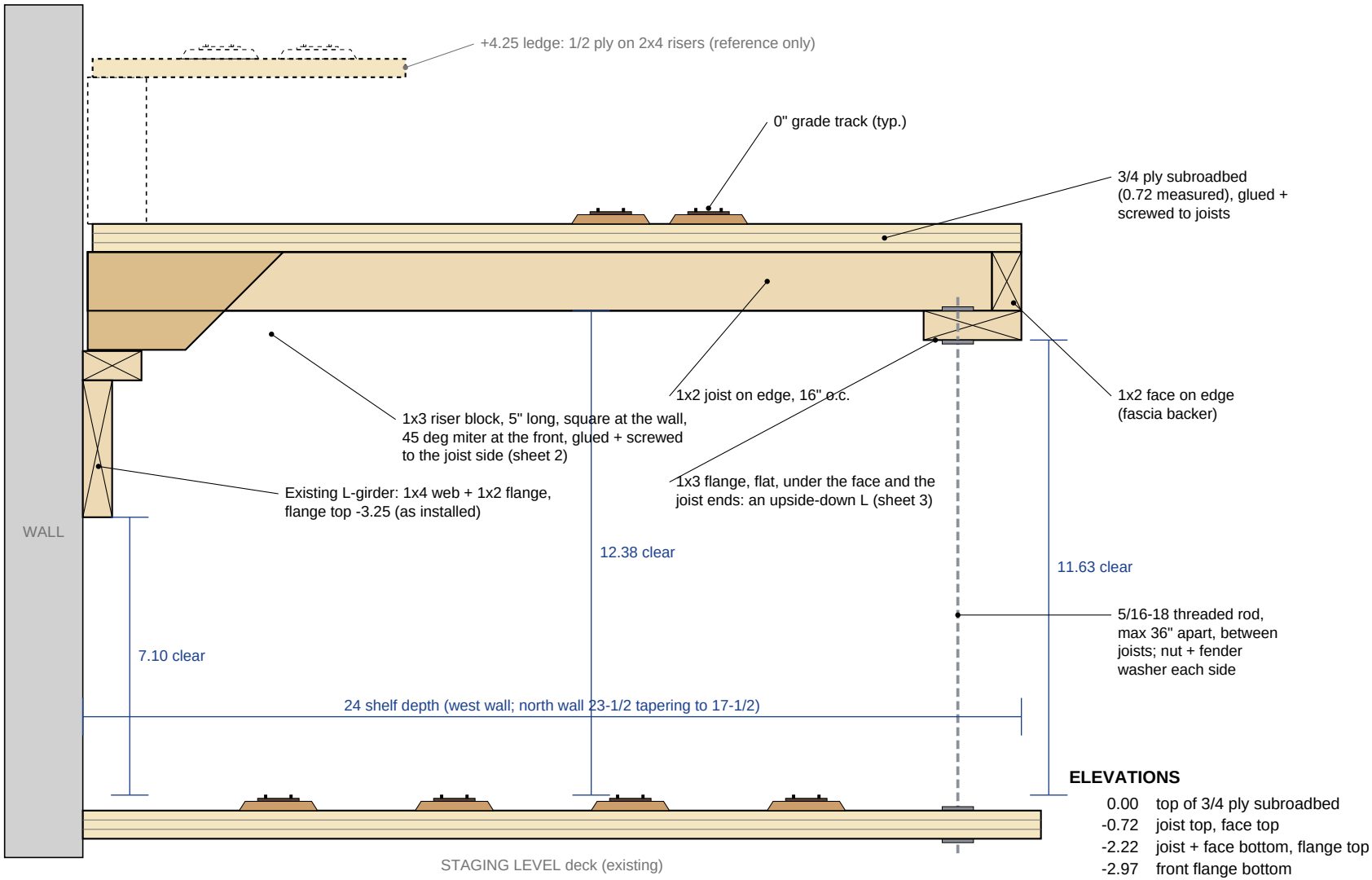
Typical shelf section - north & west walls

Section through a joist, perpendicular to the wall.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



ELEVATIONS

0.00	top of 3/4 ply subroadbed
-0.72	joist top, face top
-2.22	joist + face bottom, flange top
-2.97	front flange bottom
-3.22	riser block bottom
-3.25	girder flange top
-7.50	girder web bottom
-14.60	staging railhead
-15.00	staging deck top

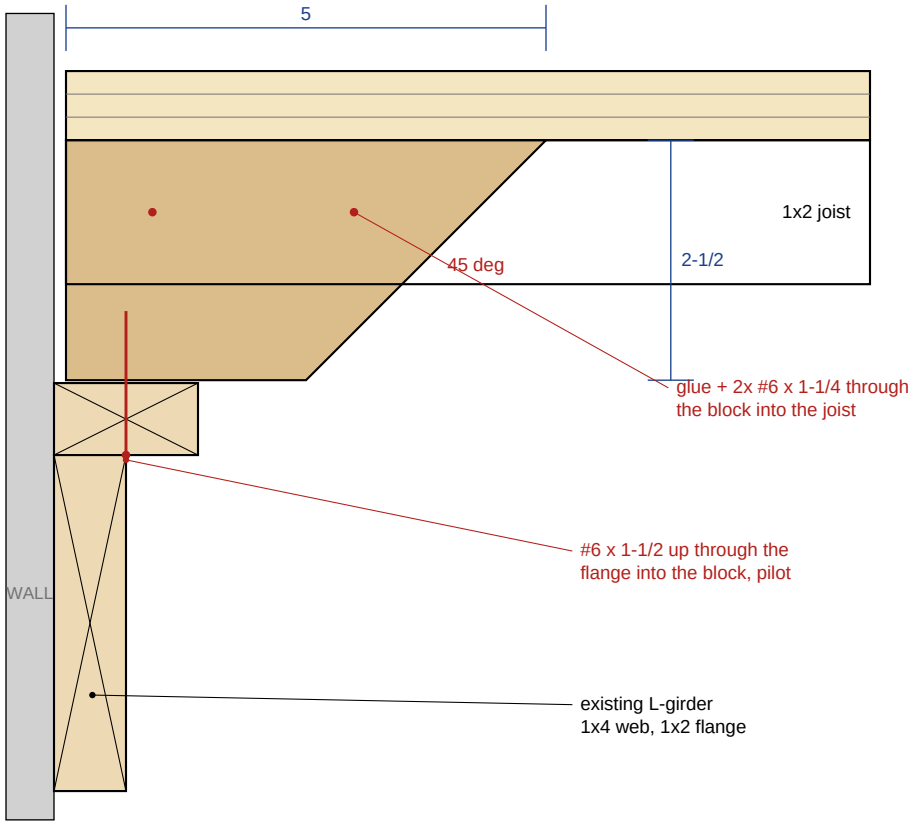
Block bottom lands 0.03 above the flange (3.25 - 2.50 - 0.72); the deck top finishes 1/32 low - within tolerance, or shim the flange.
Existing girder against the wall (web to the studs, flange on top). Track = 0.236 cork + 0.160 Atlas code 83 (caliper before clearance sign-off).
Scale 1:4 on Letter landscape. Lumber cut in section shows an X; the joist and riser block are in side view.

Details - riser block at the back girder

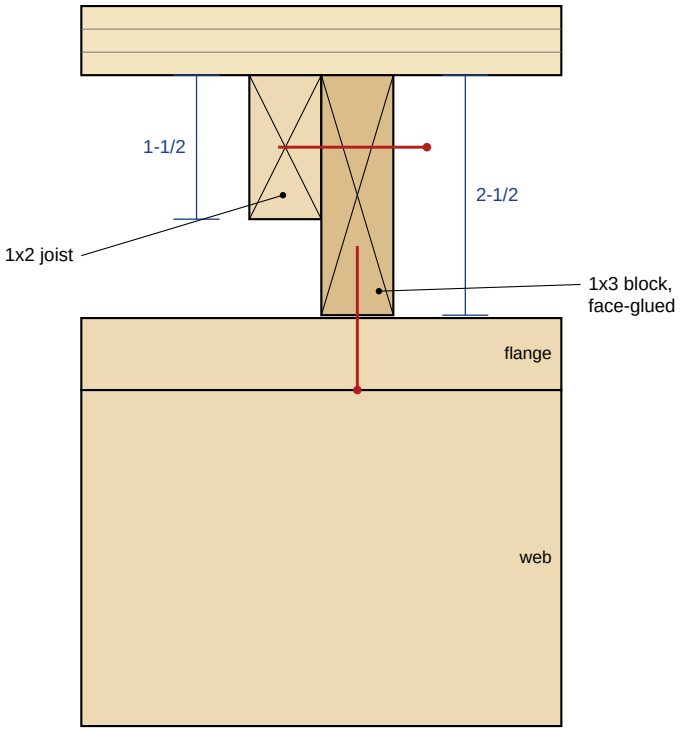
A: side view along the joist. B: looking along the joist (joist and block cut, girder in elevation).

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

A SIDE VIEW ALONG THE JOIST 1:2



B LOOKING ALONG THE JOIST 1:2



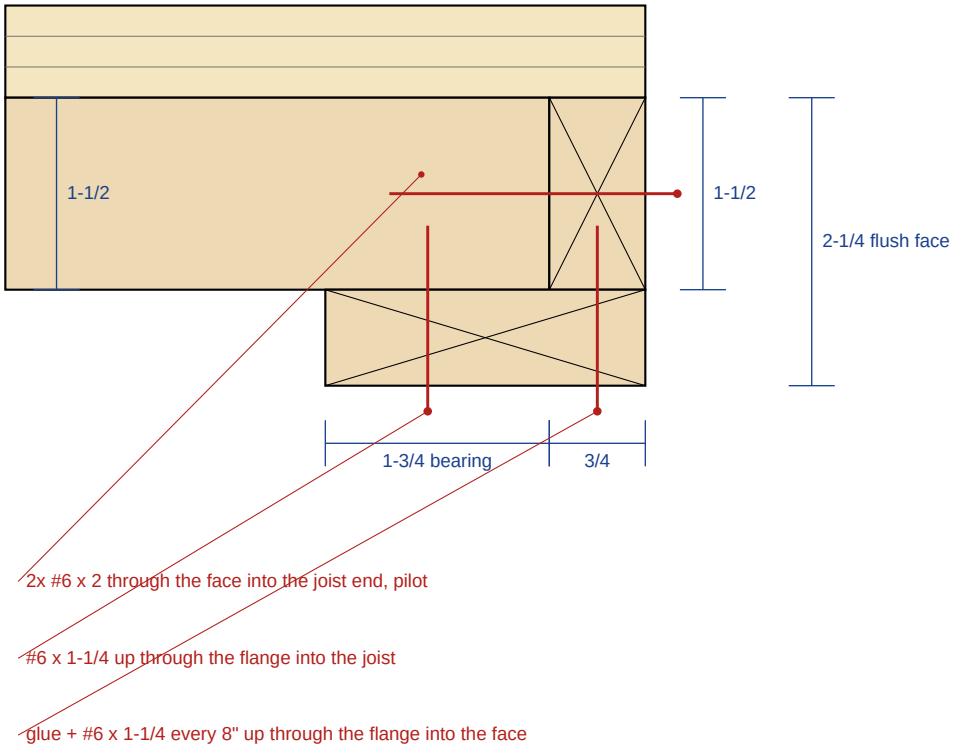
Riser block: cut from 1x3, grain running with the joist. Top flush with the joist top; the bottom sits on the flange. The wall end stays square; one full-height 45 deg miter on the front end leaves 2-1/2 of bottom, so the block still bears on the whole 1-1/2 flange. Joist and block ends sit 1/8 off the wall. Keep every block on the same side of its joist. Full-size pattern on sheet 4.

Details - front 1x2 face on a 1x3 flange

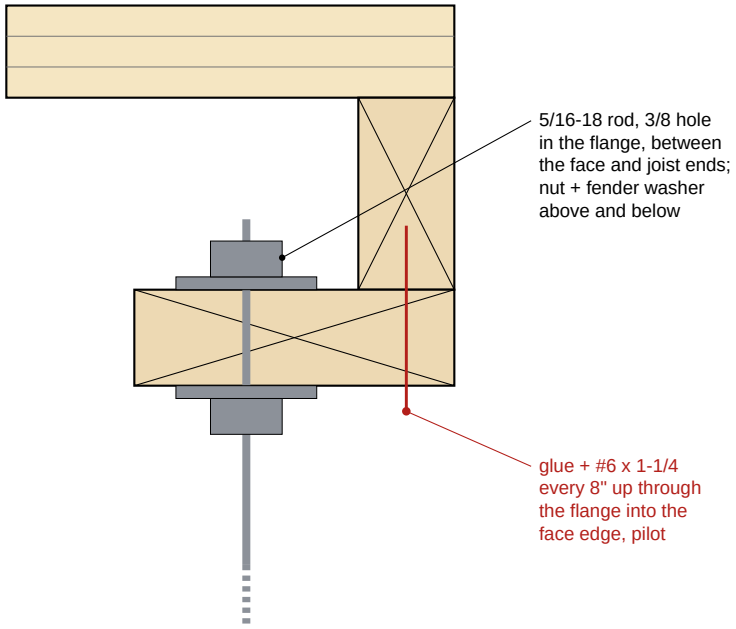
C: side view along a joist at the front edge. D: the same edge between joists, at a rod.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

C AT A JOIST 1:1.5



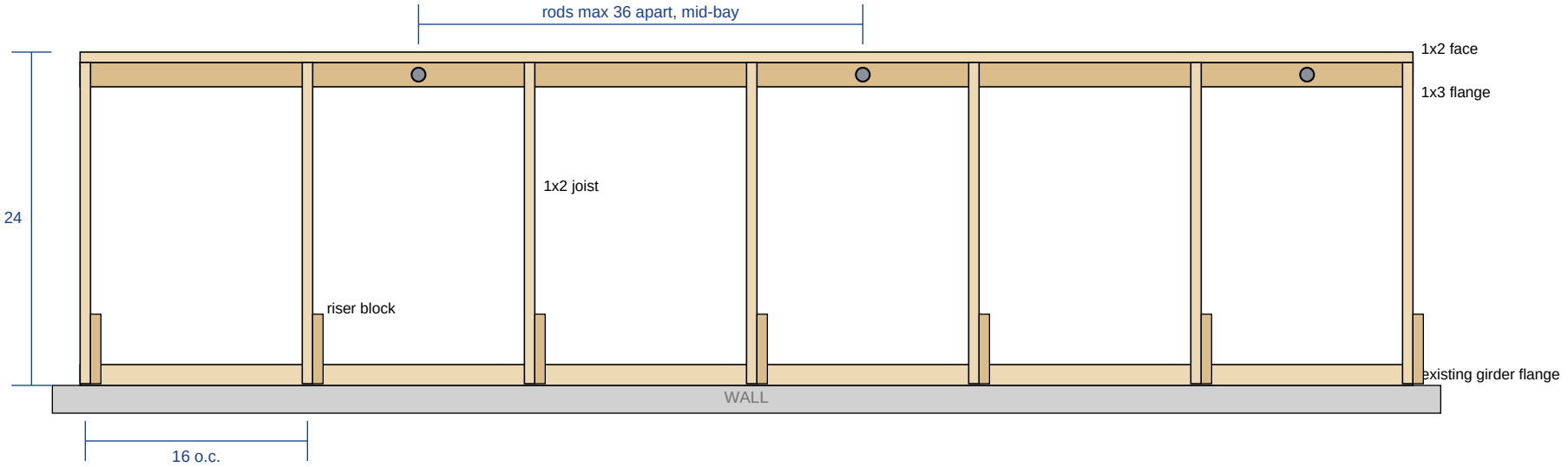
D BETWEEN JOISTS, AT A ROD 1:1.5



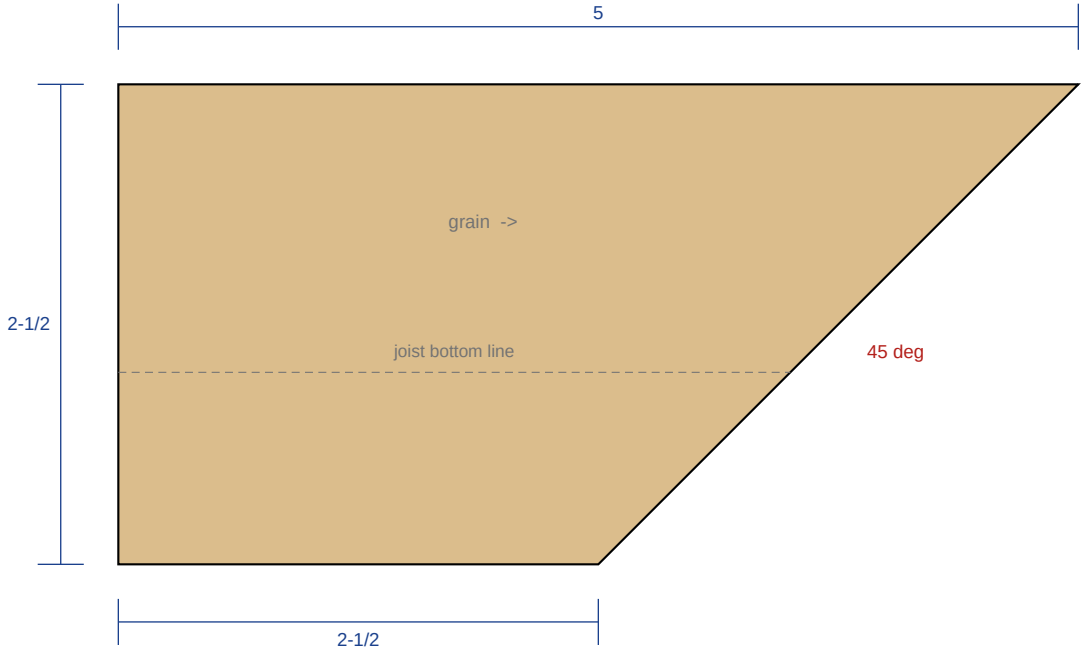
Build the front on the bench: glue the 1x2 on edge onto the 1x3 laid flat, outer faces flush - an L-girder upside down. The joists sit on the 1-3/4 of flange inside the face.
No gap: the fascia gets one flush 2-1/4 face. Face-to-flange is a long-grain glue joint; the screws clamp it while it cures. Stagger the face and flange splices.

Plan - typical 8 ft bay, deck removed + riser block pattern

Looking down. Joists at 16" o.c. land on the existing girder via the riser blocks; rods carry the front flange between joists. Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



RISER BLOCK - FULL SIZE PATTERN (1:1)



square end goes to the wall

CUT LIST PER 8 FT BAY (24" deep)

1x2 joists	7 @ 23-1/8	24 - 3/4 face - 1/8 wall gap
1x3 riser blocks	7 @ 5	square at wall, full 45 deg front
1x2 face	96	on edge, on the flange
1x3 flange	96	flat, flush with the face
5/16-18 rod	3 @ ~13	fit to the staging deck
3/4 ply	24 x 96	one rip per bay

Splice the 1x2 face and the 1x3 flange at different joists; butt joints in the 3/4 ply land on a joist. North wall: 23-1/2 deep tapering to 17-1/2 - cut each joist to depth - 3/4 - 1/8.
Scales: plan 1:12, pattern 1:1 - print at 100% and check the 5" dimension with a rule before tracing.

Assembly sequence and fastener schedule

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

1. Cut the front members: 1x2 face and 1x3 flange to length. Stagger splices so they never fall at the same joist.

2. On the bench, glue the 1x2 on edge onto the 1x3 laid flat, outer faces flush (an upside-down L-girder). Screw up through the flange into the face every 8". Let it cure.

3. Cut the riser blocks: 5" of 1x3, square at the wall end, one full-height 45 deg miter on the front end. Trace the full-size pattern on sheet 4 or make a stop block on the miter saw.

4. Glue + screw a block to the side of each joist: top flush, square end flush with the joist's wall end. Same side on every joist.

5. Prop the front assembly at height (flange top at -2.22). Drill 3/8 holes in the flange mid-bay for the rods, max 36" apart, between joists.

6. Set the joists at 16" o.c.: blocks on the flange, front ends on the flange, butted to the face. Screw up through the flange into each block, through the face into the joist end, and up through the flange into the joist.

7. Run the rods from the staging deck to the flange. Level the front edge with the nuts, then lock with the opposing nuts.

8. Dry-lay the 3/4 ply; butt joints on joists. Glue + screw every 8" along each joist, keeping screws out from under the track centrelines (feeders).

9. Check the deck top against the helix exit and the +4.25 ledge risers before the glue sets.

FASTENERS

Block to joist	glue + 2x #6 x 1-1/4
Flange up into block	#6 x 1-1/2, pilot
Flange up into face	glue + #6 x 1-1/4 @ 8"
Face into joist end	2x #6 x 2, pilot
Flange up into joist	#6 x 1-1/4
Ply to joists	glue + #6 x 1-1/4 @ 8"
Rod at flange / deck	5/16 nut + fender washer each side, 3/8 hole

CLEARANCE OVER STAGING

Under joists	12.38 in
Under front flange	11.63 in
Under girder web	7.10 in

Railhead = staging deck + 0.236 cork + 0.160 track.